# Solve with us 3.2 - Not Graded

This assignment will not be graded and is only for practice.
Key points: 1Key points: 1

- Definition of linear dependence: A set of vectors $v_1, v_2, ...v_n \in V$ is said to be linearly dependent if there exist scalars $a_1, a_2, ...a_n \in R$, not all zero such that $a_1 v_1 + a_2 v_2 + ... + a_n v_n = 0$.
- To check the vectors $v_1, v_2, ...v_n \in R^m$ (with usual addition and scalar multiplication) are linearly dependent, we have to verify that the homogeneous system of linear equations $Ax = 0$ has infinitely many solutions, where the $j^{th}$ column of $A$ is $v_j$.
- Definition of linear independence: A set of vectors $v_1, v_2, ...v_n \in V$ is said to be linearly independent if $a_1 v_1 + a_2 v_2 + ... + a_n v_n = 0$ implies that $a_i = 0$ for $i = 1, 2, ...n$.
- To check the vectors $v_1, v_2, ...v_n \in R^m$ (with usual addition and scalar multiplication) are linearly independent, we have to verify that the homogeneous system of linear equations $Ax = 0$ has only the trivial solution (i.e., $x = 0$), where the $j^{th}$ column of $A$ is $v_j$.
- If $S$ is a set containing zero vector, then the set $S$ is a linearly dependent set.
- Let $S$ be a set of vectors from a vector space $V$. If an element of $S$ is a scalar multiple of another element from the set $S$, then $S$ is a linearly dependent set.

*1 point*
Vectors (2,1) and (6, 3) from $R^2$ are

◯ linearly independent.
◯ linearly dependent.
◯ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
linearly dependent.

*1 point*
Suppose the set $\{(3, 7, 4), v\}$ is a linearly dependent set in the vector space $R^3$ with usual addition and scalar multiplication. Which of the following vectors are possible candidates for $v$?

☐
$(0, 0, 0)$
☐
$(5, \frac{35}{3}, \frac{20}{3})$
☐
$(1, 0, 0)$
☐
$(6, 14, -8)$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$(0, 0, 0)$
$(5, \frac{35}{3}, \frac{20}{3})$

*1 point*
The set of vectors $\{(1, 1, -1), (-1, 1, 1), \text{ and } (0, \frac{1}{2}, 0)\}$ in [illegible]s

◯ linearly independent.

○ linearly dependent.
○ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
linearly dependent.

*1 point*

$M_{2\times 2}(R)$M2×2(R) denotes the vector space consisting of real square matrices of order 2 with usual matrix addition and scalar multiplication. Consider the following matrices:

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, M_2 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, M_3 = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$ M1 = [100−1] , M2 = [10−10] , M3 = [1−101]

Which of the following options is(are) true?

[Hint:Hint: Form a system of linear equations using matrix equation $x\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} + y\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$x [a11a21 a12a22] + y [b11b21b12b22] = 0]

☐ $\{M_1, M_2\}$\{M1, M2\} is a linearly independent set.
☐ $\{M_1, M_2, M_3\}$\{M1, M2, M3\} is a linearly dependent set.
☐ $\{M_2, M_3\}$\{M2, M3\} is a linearly dependent set.
☐ $\{M_1, M_3\}$\{M1, M3\} is a linearly independent set.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{M_1, M_2\}$\{M1, M2\} is a linearly independent set.
$\{M_1, M_3\}$\{M1, M3\} is a linearly independent set.

Key points: 2Key points: 2

- Given two vectors $(x_1, y_1)$, and $(x_2, y_2) \in R^2$ (x1, y1), and (x2, y2) ∈ R2 construct $A = \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix}$ A = [x1y1x2y2]. Then these vectors are linearly dependent if and only if $det(A) = 0$det(A) = 0 and linearly independent if and only if $det(A) \neq 0$det(A)  = 0.
- Given three vectors $(x_1, y_1, z_1), (x_2, y_2, z_2)$ and $(x_3, y_3, z_3) \in R^3$ (x1, y1, z1), (x2, y2, z2 ) and (x3, y3, z3) ∈ R3 construct $A = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$A = [x1y1z1x2y2z2x3y3z3]. Then these vectors are linearly dependent if and only if $det(A) = 0$det(A) = 0 and linearly independent if and only if $det(A) \neq 0$det(A)  = 0

*1 point*

Vectors (2,3) and (4, 5) from $R^2$ R2 are

[Hint:Hint: Form the matrix using the vectors and calculate the determinant of the matrix.]

- ◯ linearly independent.
- ◯ linearly dependent.
- ◯ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
linearly independent.

*1 point*

Consider a set of vectors $S = \{(1,2),(-1,2),(3,1),(-3,1)\} \subset$ S = {(1, 2), (−1, 2), (3, 1), (−3, 1)} ⊂ $R^2$ R2. Choose the set of correct options.

[Hint:Hint: Form the matrix using the vectors as the columns for each option and calculate the determinant of the matrix.]

- [ ] $\{(1,2),(-1,2)\}$ {(1, 2), (−1, 2)} is a linearly dependent set.
- [ ] $\{(-1,2),(-3,1)\}$ {(−1, 2), (−3, 1)} is a linearly independent set.
- [ ] $\{(-3,1),(-1,2)\}$ {(−3, 1), (−1, 2)} is a linearly independent set.
- [ ] $\{(3,1),(-3,1)\}$ {(3, 1), (−3, 1)} is a linearly dependent set.
- [ ] $\{(3,1),(1,2)\}$ {(3, 1), (1, 2)} is a linearly independent set.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$\{(-1,2),(-3,1)\}$ {(−1, 2), (−3, 1)} is a linearly independent set.
$\{(-3,1),(-1,2)\}$ {(−3, 1), (−1, 2)} is a linearly independent set.
$\{(3,1),(1,2)\}$ {(3, 1), (1, 2)} is a linearly independent set.

*1 point*

The set of vectors $\{(1,1,0),(0,1,1),(1,0,1)\}$ {(1, 1, 0), (0, 1, 1), (1, 0, 1)} is

[Hint:Hint: Form the matrix using the vectors as the columns and calculate the determinant of the matrix.]

- ◯ linearly dependent.
- ◯ linearly independent.
- ◯ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
linearly independent.

*1 point*

Suppose $\{(2,3,0),(0,1,0),v\}$ {(2, 3, 0), (0, 1, 0), v} is a linearly dependent set in the vector space $R^3$ R3 with usual addition and scalar multiplication. Which of the following vectors are possible candidates for $v$ v?

- [ ] $(2,3,1)$ (2, 3, 1)
- [ ] $(1,2,0)$ (1, 2, 0)
- [ ] $(1,3,0)$ (1, 3, 0)

☐ $(0, 1, 1)$

☐ $(\pi, e, 0)$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$(1, 2, 0)$

$(1, 3, 0)$

$(\pi, e, 0)$

*1 point*

Consider three vectors $v_1 = (1, 0, 0), v_2 = (1, 1, 0), v_3 = (1, 1, 1)$ in the vector space $R^3$, with usual addition and scalar multiplication. Which of the following option is true?

[Hint: Find out the vectors of the given sets in the options and form the matrices using the vectors as the columns and calculate the determinant of the matrices.]

○ $\{v_1, v_2, v_3\}$ is a linearly dependent set.

○ $\{v_1 + v_2, v_1 + v_3, v_3\}$ is a linearly dependent set.

○ $\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$\{v_2 - v_1, v_1, v_2\}$ is a linearly dependent set.

Key points: 3

- Any set of $n$ vectors in $R^2$ with $n \geq 3$ is linearly dependent.
- Any set of $n$ vectors in $R^3$ with $n \geq 4$ is linearly dependent.
- Any set of $n + 1$ vectors in $R^n$ is linearly dependent.
- Any set containing a linearly dependent set is linearly dependent.
- If a set is linearly independent, then every subset of it, is linearly independent.

*1 point*

The subset $S = \{(3, 1), (1, 2), (-3, 1)\}$ of $R^2$ is

[Hint: Any set of $n$ vectors in $R^2$ with $n \geq 3$ is linearly dependent.]

○ linearly dependent.

○ linearly independent.

○ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

linearly dependent.

*1 point*

Set $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$ subset of $R^3$ is

[Hint:Hint: Any set of $n$ vectors in $R^3$ with $n \geq 4$ are linearly dependent.]

- ○ linearly dependent.
- ○ linearly independent.
- ○ None of the above.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
linearly dependent.

*1 point*

Let $S$ be a linearly dependent subset of $R^3$. Which of the following options are true?

- ☐ $S \cup \{(1,0,0)\}$ must be linearly dependent.
- ☐ $S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.
- ☐ $S \setminus \{v\}$ must be linearly dependent for any $v \in S$.
- ☐ There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$S \cup \{(1,0,0)\}$ must be linearly dependent.
$S \cup \{v\}$ must be linearly dependent for any $v \in R^3$.
There may exist some $v \in S$, such that $S \setminus \{v\}$ is still linearly dependent.

Check Answers

Your score is: 0/12